KRISH JOSHI

COMPUTER SCIENCE UNDERGRADUATE (3RD YEAR) • AI SYSTEMS & SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Computer Science 3rd-year undergraduate (Pre-Final Year) with demonstrated production experience engineering autonomous AI harnesses, containerized systems environments, and deterministic evaluation pipelines. Authored 10 terminal-based systems environments in Docker (Harbor format) under Handshake's Project Dynamo, measuring frontier model code across distributed DAGs, GPU compute, and Linux recovery. Audited 32 multi-turn agent interaction trajectories under Project Octahedron, ensuring strict data provenance, tool grounding, and error recovery. Proficient in Python, C/C++, TypeScript, Docker, SQL, and Linux OS internals with Google Cybersecurity Professional certification.

EDUCATION & ACADEMICS

Bachelor of Technology (B.Tech), Computer Science & Engineering — **Expected Graduation: 2027/2028**

Institute of Advanced Research (IAR) — Gandhinagar, Gujarat, India — Status: Active 3rd-Year Undergraduate (Pre-Final Year)

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS (SQL), Computer Networks, Software Engineering

Certifications: Google Cybersecurity Professional Certificate | Introduction to Networking (NVIDIA)

TECHNICAL SKILLS

AI Systems & Agents:	Agent Harness Engineering, Tool-Call Grounding, Retrieval Provenance, Prompt Hardening, Harbor Format, Hallucination Auditing
Software & Languages:	Python (FastAPI, PyTest, NumPy), C/C++, TypeScript / JavaScript, Bash Scripting, SQL (PostgreSQL, SQLite)
Infrastructure & DevOps:	Docker Containerization, Linux CLI & System Calls, Git/GitHub Actions CI/CD, Sandboxing (cgroups, quotas), CTRF Schemas
Systems & Security:	Application Security (VAPT basics), Access Control & Token Auth, Distributed State DAGs, Concurrency & Memory Fences

WORK EXPERIENCE (INDUSTRY CONTRACTS)

AI Systems & Test Harness Developer (Independent Contractor) — **July 2026 – September 2026**

Handshake — AI Evaluation Program (Project Dynamo) — Remote (Contract Completed)

- **Containerized Benchmarks:** Authored 10 terminal-based engineering environments in Docker evaluating autonomous agents across distributed systems, GPU compute, and automated system recovery.
- **Deterministic Grading Suites:** Built PyTest grading suites with CTRF schema reporting and SHA-256 state assertions; verified reference solutions passed 100% while frontier models failed on ≥60% of initial baseline runs.
- **Pass@k Calibration:** Calibrated task difficulty across empirical pass@2 and pass@5 trial regimes, eliminating superficial string-matching shortcuts and forcing deep multi-step algorithmic reasoning.
- **Runtime Sandboxing:** Hardened air-gapped evaluation containers with strict path-traversal assertions (assert_safe_path) and memory/CPU quotas, preventing environment tampering and ensuring test reproducibility.

AI Alignment & Workflow Specialist (Independent Contractor) — **September 2026 – Present**

Handshake — Enterprise Model Comparison Program (Project Octahedron) — Remote (Active Contract alongside Studies)

- **Trajectory Auditing:** Evaluated 32 multi-turn agent interaction transcripts totaling 27+ hours of active review, ensuring strict factual grounding, API contract adherence, and data provenance across real-world tools.
- **Rubric Verification:** Audited model outputs against stringent multi-dimensional rubrics (instruction following, technical accuracy, safety), detecting subtle hallucinations and non-deterministic tool-call discrepancies.
- **Evidentiary Precision:** Maintained 100% compliance with client anti-AI styling and evidentiary standards, submitting zero-defect rationales with exact mathematical alignment to rating scores.

SELECTED SYSTEMS & AI PROJECTS

Distributed Execution Graph & DAG State Reconciliation — Python, SQLite, Graph Algorithms, PyTest, JSONL

- Architected a systems pipeline requiring AI models to parse 3,500+ line cluster execution logs and reconstruct valid state DAGs across branching rollbacks, node crashes, and data dependencies.
- Graded solutions on recursive cycle detection, tree pruning, and deterministic tie-breaking with ±1.0s wall-clock accuracy, eliminating regex shortcuts.

W4A16 Quantized GEMM GPU Kernel Benchmark — OpenCL C, C/C++, Python, PyTest, SIMT Concurrency

- Engineered a matrix multiplication benchmark running tiled dot products with asymmetric 4-bit packed weights and FP32 accumulation in OpenCL; eliminated workgroup races using local memory barrier fences.
- Automated verification against FP64 reference baselines across non-power-of-two matrix shapes, catching ragged edge boundary bugs in agent-generated code.

Octahedral Rotation Group 3D Constraint Solver — Python, NumPy, Discrete Math, CSP Backtracking

- Designed a combinatorial benchmark modeling 3D spatial constraints over the 24-element octahedral group using non-commutative composition paths, forward checking, and matrix trace parity verification.